Grounded CAD: Geometry Whispers for CAD Edit

Team Night Owls - IDETC 2026 Problem Statement 3

Final presentation PDF for repository submission

Goal

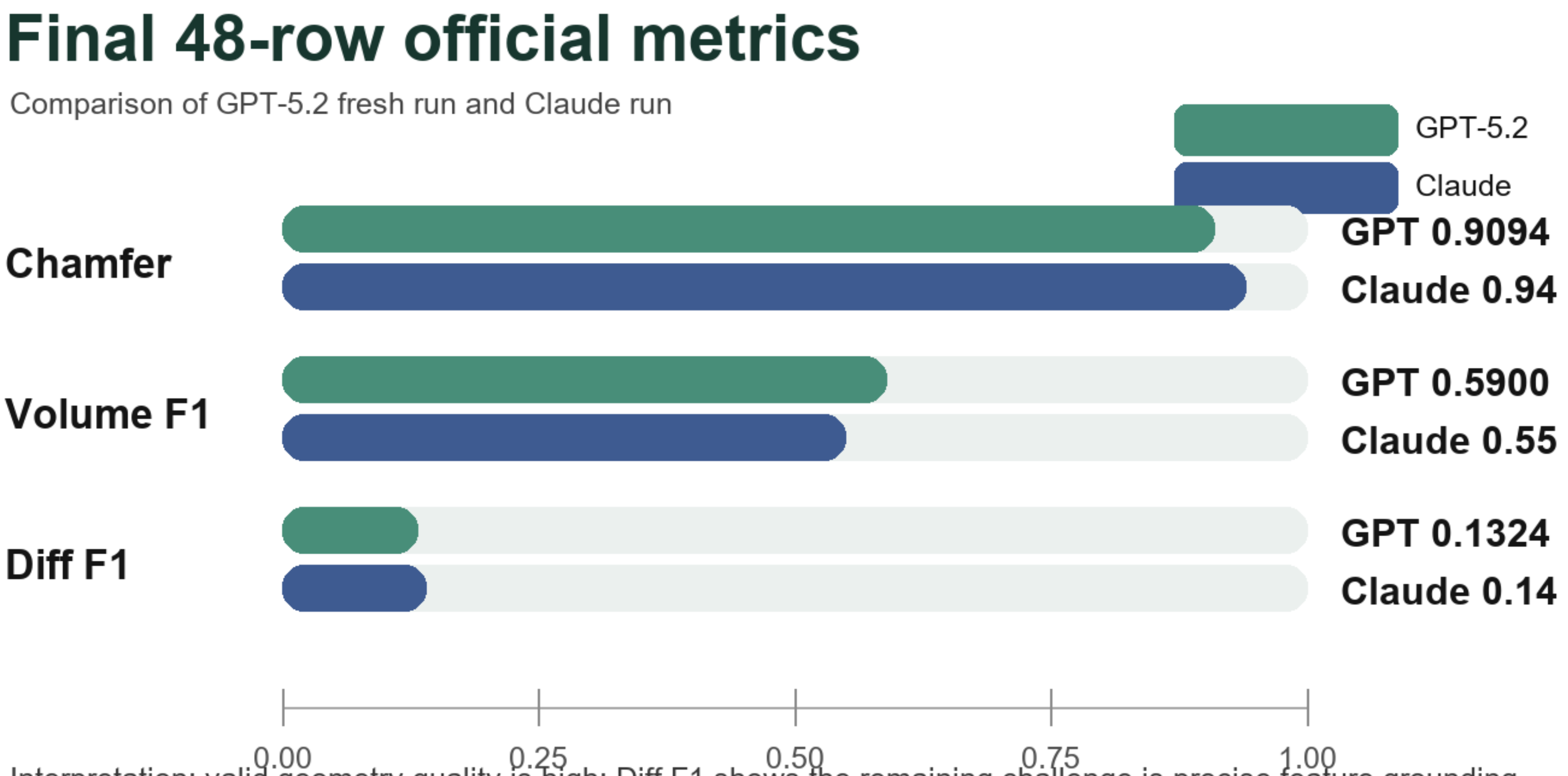
Build a harness that edits existing STEP models from natural-language instructions while maximizing automatic CAD-edit metrics.

Method overview

Grounder	Extracts a compact B-Rep census from the STEP model: solids, faces, holes, rims, dimensions, bounding boxes, and candidate edit sites.
Classifier	Parses the instruction into operation type, target feature, dimensions, direction, and constraints.
Planner	Chooses deterministic local CAD tools first; falls back to constrained CadQuery generation when necessary.
Executor	Runs CadQuery/OpenCascade edits in a sandboxed child process with watchdog protection.
Verifier/Critic	Checks validity, geometric change, locality, and failure buckets such as identity, wrong scope, misclassification, or kernel failure.

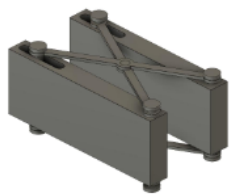
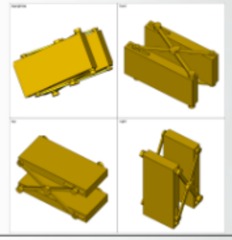
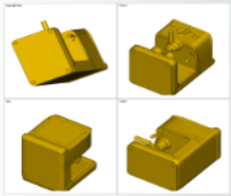
Design principle: avoid direct STEP text edits; use geometry-aware B-Rep operations plus validation.

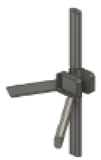
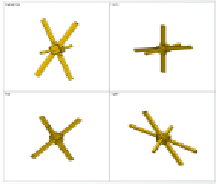
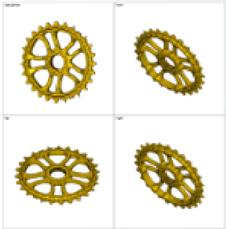
Final scores

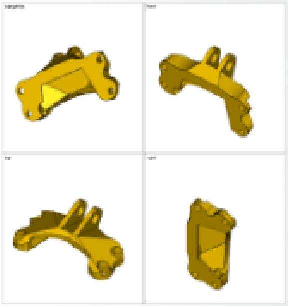


Interpretation: valid geometry quality is high; Diff F1 shows the remaining challenge is precise feature grounding.

Qualitative evaluation: five selected request IDs

Initial Model	Grounded CAD	Scores
		Diff F1 0.023 Vol F1 0.361 Chamfer 0.975
		Diff F1 0.245 Vol F1 0.638 Chamfer 0.978

Initial Model	Grounded CAD	Scores
Add third rotor blade to assembly 		Diff F1 0.108 Vol F1 0.347 Chamfer 0.985
		Diff F1 0.143 Vol F1 0.912 Chamfer 0.973

Initial Model	Grounded CAD	
		Diff F1 0.082 Vol F1 0.779 Chamfer 0.982

Takeaways and next steps

- Valid STEP export alone is not enough; Diff F1 depends on changing the requested local feature.
- The strongest signals came from STEP-derived B-Rep facts, true orthographic views, candidate enumeration, and failure-aware validation.
- Remaining score loss is dominated by wrong edge/scope, misclassification, identity outputs, and OpenCascade boolean/rendering failures.
- Next improvements should confidence-gate deterministic operations and re-ground alternate candidates before accepting wrong-scope edits.